Force on a Charge and a Thought Experiment

From electromagnetics, the force on a charge q moving with velocity $\mathbf{v}$ in the presence of electric and magnetic fields is

$$\boxed{\mathbf{F} = q\mathbf{E} + q\,\mathbf{v} \times \mathbf{B}} \tag{1}$$

This is called the **Lorentz force law**. It tells us that

- the electric field can exert force even on a charge at rest,
- the magnetic field exerts force only when the charge is moving.

Immediate conclusion from (1)**:**

$$\boxed{\text{If } \mathbf{v} = 0, \text{ then the magnetic force is zero.}}$$

So a magnetic field alone cannot push a stationary charge.

Thought experiment

Consider a uniform magnetic field

$$\mathbf{B} = B_0\,\hat{\mathbf{z}},$$

and two charges q and q', both at rest in the laboratory frame.

Force in the laboratory frame

In the laboratory frame S, both charges are at rest. Therefore,

$$\mathbf{v} = 0.$$

Hence the magnetic part of the Lorentz force vanishes:

$$q\,\mathbf{v} \times \mathbf{B} = 0.$$

So the force on q is only due to the electric field created by q':

$$\boxed{\mathbf{F} = q\mathbf{E}} \tag{2}$$

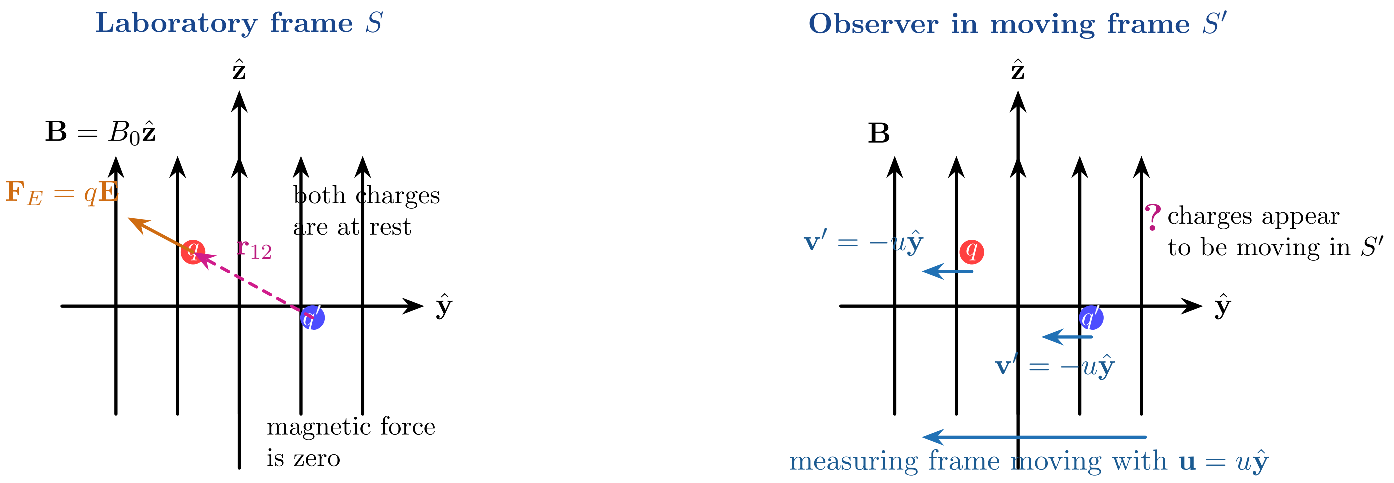


Figure 1: Thought experiment: in the laboratory frame both charges are at rest, so the magnetic force is zero. For an observer moving with velocity $\mathbf{u}$, the charges appear to move. This raises a deep question: does the measured force change?

If $\mathbf{r}_{12}$ is the vector from q' to q, then

$$\mathbf{E}_{q'}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q'\,\mathbf{r}_{12}}{r_{12}^3}. \tag{3}$$

Therefore, the force on q due to q' is

$$\boxed{\mathbf{F}_q = \frac{qq'}{4\pi\varepsilon_0}\frac{\mathbf{r}_{12}}{r_{12}^3}.} \tag{4}$$

Now consider a moving observer

Suppose an experimentalist measures the force from a vehicle moving with velocity

$$\mathbf{u} = u\,\hat{\mathbf{y}}.$$

Then, in that moving frame S',

- the observer sees the charges as moving,
- so one may wonder whether a magnetic-force contribution should now appear.

Indeed, in that moving frame, the charges are no longer at rest, so the term

$$q\,\mathbf{v}' \times \mathbf{B}'$$

need not obviously vanish.

This leads to a very important conceptual question:

Question to think about: If a different observer sees the charges moving, will the measured force on the charge be different?

$$\boxed{\text{Does a moving measuring instrument give a different force reading?}}$$

The actual physical force on the charge should not become a different "real" force merely because the observer is moving. So how should the electric and magnetic descriptions adjust themselves from one frame to another?

Force in the moving frame

In the moving frame S', the charge velocity is

$$\mathbf{v}' = -\mathbf{u}.$$

Hence the Lorentz force on q becomes

$$\mathbf{F}' = q\mathbf{E}' + q\,\mathbf{v}' \times \mathbf{B}' = q\mathbf{E}' - q\,(\mathbf{u} \times \mathbf{B}'). \tag{5}$$

Now, the same physical interaction is being described by two inertial observers. So the description must be *consistent*. At the level of this simple low-speed thought experiment, we therefore write

$$\mathbf{F}' \approx \mathbf{F}. \tag{6}$$

Using (2) and (5),

$$q\mathbf{E} \approx q\mathbf{E}' - q(\mathbf{u} \times \mathbf{B}').$$

Cancelling q,

$$\mathbf{E} \approx \mathbf{E}' - \mathbf{u} \times \mathbf{B}'.$$

Hence,

$$\boxed{\mathbf{E}' \approx \mathbf{E} + \mathbf{u} \times \mathbf{B}'.} \tag{7}$$

If the observer speed is small enough that the difference between $\mathbf{B}'$ and $\mathbf{B}$ can be neglected to first order, then

$$\boxed{\mathbf{E}' \approx \mathbf{E} + \mathbf{u} \times \mathbf{B}.} \tag{8}$$

Main conclusion of the thought experiment: A field that appears purely electric in one frame need not remain purely electric in another frame.
To first order in relative speed,

$$\boxed{\mathbf{E}' \approx \mathbf{E} + \mathbf{u} \times \mathbf{B}.}$$

So electric and magnetic fields are not completely separate entities; they are different aspects of the same electromagnetic reality.

Important correction / interpretation

It is tempting to say:

"the electric field has modified itself"

but the more precise statement is:

Better interpretation: The *electric and magnetic field components seen by the observer* are different in different inertial frames.
It is not that the actual physics changes. Rather, the decomposition into "electric part" and "magnetic part" depends on the frame of observation.

Also, in the moving frame the source charge q' is itself moving. So the electric field seen in that frame need not have the simple static Coulomb form.

Final summary of this section

1. In the laboratory frame, the charges are at rest, so magnetic force is zero.
2. In a frame moving with velocity $\mathbf{u}$, the charges appear to move with velocity $-\mathbf{u}$.
3. The force there is
$$\mathbf{F}' = q\mathbf{E}' - q(\mathbf{u} \times \mathbf{B}').$$
4. Consistency of the description leads, at low speed, to
$$\boxed{\mathbf{E}' \approx \mathbf{E} + \mathbf{u} \times \mathbf{B}.}$$

This is one of the first clues that electric and magnetic fields are fundamentally connected.

Remark. A fully exact derivation of the relation between $\mathbf{E}$ and $\mathbf{B}$ in different frames requires **special relativity**. The present discussion should be viewed as a clear physical motivation rather than the full relativistic proof.

From the Moving-Frame Discussion to EMF

In the previous discussion, we were led to the low-speed relation

$$\boxed{\mathbf{E}' \approx \mathbf{E} + \mathbf{u} \times \mathbf{B}} \tag{9}$$

which tells us that, for an observer moving with velocity $\mathbf{u}$, the electric field they perceive is modified by the magnetic field.

Key idea for the next step: If a *conductor* moves through a magnetic field with velocity $\mathbf{u}$, then the charges inside the conductor move with the conductor. Hence those charges experience the magnetic force

$$q\,\mathbf{u} \times \mathbf{B}.$$

This is the starting point for understanding **motional EMF**.

Motional Force on Charges Inside a Moving Conductor

Consider a straight conducting rod of length ℓ, moving with constant velocity

$$\mathbf{u} = u\,\hat{\mathbf{x}}$$

through a uniform magnetic field

$$\mathbf{B} = -B\,\hat{\mathbf{z}},$$

that is, the magnetic field is *into the page*.

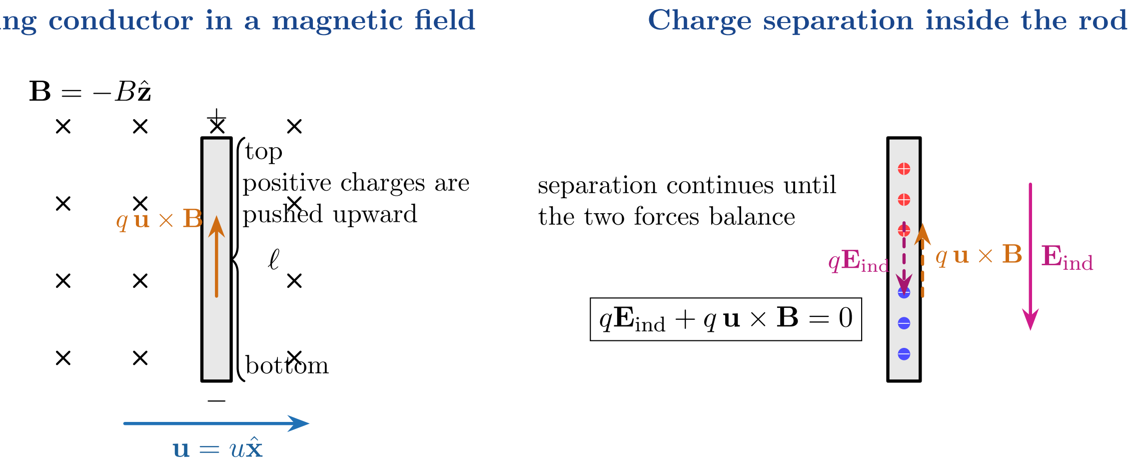


Figure 2: A conductor moving through a magnetic field. The magnetic force separates charges inside the rod, which sets up an internal electric field. At equilibrium, this electric field balances the magnetic force.

For a positive charge inside the rod,

$$\mathbf{F} = q\mathbf{E} + q\,\mathbf{u} \times \mathbf{B}. \tag{10}$$

Here the electric field $\mathbf{E}$ is the field created by the separated charges inside the conductor. At equilibrium, the charges stop redistributing, so the net force along the rod becomes zero:

$$q\mathbf{E}_{\text{ind}} + q\,\mathbf{u} \times \mathbf{B} = 0. \tag{11}$$

Therefore,

$$\boxed{\mathbf{E}_{\text{ind}} = -\,\mathbf{u} \times \mathbf{B}.} \tag{12}$$

Main physical meaning: A magnetic field does not directly create a static potential difference by itself. Rather, when the conductor *moves*, the magnetic force separates charges, and this charge separation builds up an electric field inside the conductor.
So motional EMF ultimately appears as a consequence of:

$$\boxed{q\,\mathbf{u} \times \mathbf{B} \quad \Longrightarrow \quad \text{charge separation} \quad \Longrightarrow \quad \mathbf{E}_{\text{ind}}.}$$

Potential Difference Across the Rod

If the rod has length ℓ and $\mathbf{u} \perp \mathbf{B}$, then the magnitude of the induced electric field is

$$E_{\text{ind}} = uB.$$

Hence the potential difference between the two ends of the rod is

$$\boxed{V = E_{\text{ind}}\,\ell = uB\ell.} \tag{13}$$

This potential difference is called the **motional EMF**. Thus,

$$\boxed{\mathcal{E} = uB\ell} \tag{14}$$

for the simple case of a straight rod moving perpendicular to the magnetic field.

Direction of polarity: The end toward which positive charges are pushed becomes at higher potential. So the polarity is determined directly from the direction of

$$\mathbf{u} \times \mathbf{B}.$$

General Meaning of EMF

The above discussion suggests a more general interpretation of EMF.

What is EMF? EMF is the *work done per unit charge* by non-electrostatic means in driving charge around a circuit.
So EMF is not merely "voltage" in the ordinary electrostatic sense. It is a more general quantity that measures how charges are driven.

For a moving conductor, the effective force per unit charge is

$$\mathbf{E} + \mathbf{u} \times \mathbf{B}.$$

Therefore, for a closed path C, the general expression for EMF is

$$\boxed{\mathcal{E} = \oint_C (\mathbf{E} + \mathbf{u} \times \mathbf{B}) \cdot d\ell.} \tag{15}$$

If the circuit is stationary, then $\mathbf{u} = 0$, and

$$\mathcal{E} = \oint_C \mathbf{E} \cdot d\ell.$$

If the EMF arises purely due to motion in a magnetic field, then

$$\boxed{\mathcal{E} = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\ell.}$$

To connect the above rod picture with circuit language, imagine the rod forming part of a circuit.

Motional EMF in a simple circuit

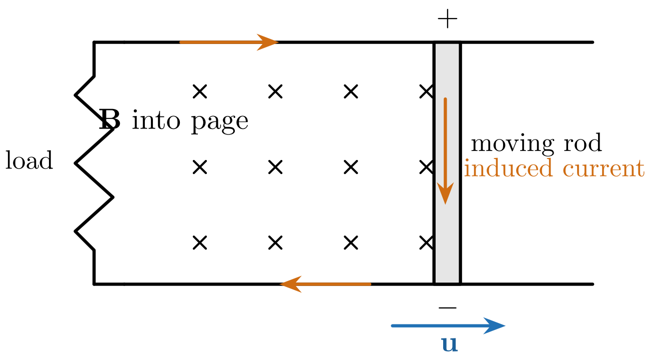


Figure 3: A moving rod as part of a closed circuit. Charge separation in the rod creates a motional EMF, which can drive current around the loop.

Summary of the Discussion

Final summary

1. A moving observer suggests the effective combination
$$\mathbf{E} + \mathbf{u} \times \mathbf{B}.$$
2. A conductor moving through a magnetic field experiences magnetic force on its charges.
3. These charges separate and create an induced internal field:
$$\boxed{\mathbf{E}_{\text{ind}} = -\,\mathbf{u} \times \mathbf{B}.}$$
4. For a rod of length ℓ,
$$\boxed{\mathcal{E} = uB\ell}$$
when $\mathbf{u} \perp \mathbf{B}$ and the rod is perpendicular to both.
5. In general,
$$\boxed{\mathcal{E} = \oint_C (\mathbf{E} + \mathbf{u} \times \mathbf{B}) \cdot d\ell.}$$

Faraday's Similar Experiment

Let us now consider another important experimental observation by Faraday.

Experiment 2

If the magnetic field through a closed conducting loop is changed with time, the galvanometer shows a deflection, indicating that a current flows in the circuit. Thus, even without directly moving the wire in all cases, a *changing magnetic field* can produce an induced current.

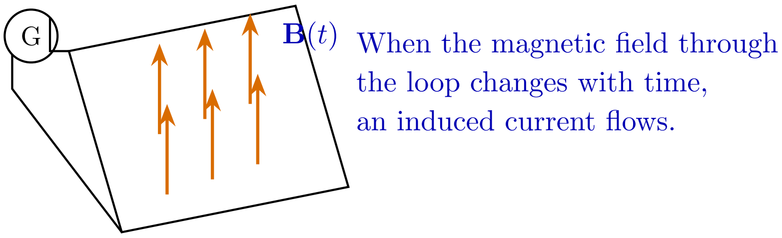


Figure 4: A changing magnetic field through a conducting loop produces an induced emf and hence a current.

Both of Faraday's experimental observations can now be summarized in a single statement:

Observation by Faraday

An induced electromotive force (emf) is created whenever the magnetic flux through a circuit changes. If the circuit is closed, this emf drives an induced current. The direction of the induced emf/current is such that it opposes the change in magnetic flux producing it.

Faraday's Law of Electromagnetic Induction

The above observation is expressed mathematically by **Faraday's law**:

$$\mathcal{E} = -\frac{d\Phi_B}{dt}, \tag{16}$$

where

$$\Phi_B = \int_S \mathbf{B} \cdot d\mathbf{S} \tag{17}$$

is the magnetic flux through the surface bounded by the circuit.

For a coil of N turns, Faraday's law becomes

$$\mathcal{E} = -N\frac{d\Phi_B}{dt}. \tag{18}$$

The negative sign indicates **Lenz's law**, i.e., the induced emf always acts in a direction that opposes the change in magnetic flux.

Key Takeaway

A current is induced in a closed circuit whenever the magnetic flux linked with it changes. Thus, *changing flux* is the essential cause of electromagnetic induction.

Understanding Lenz's law using Experiment 1

Consider again the conducting rod sliding on rails in the presence of a uniform magnetic field.

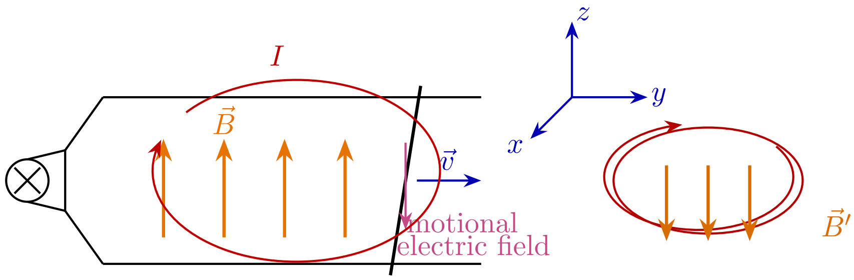


Figure 5: As the rod moves to the right, the enclosed area increases. Hence the magnetic flux due to the external field $\vec{B}$ increases. The induced current therefore flows in such a direction that the magnetic field $\vec{B}'$ produced by it opposes this increase in flux.

When the rod moves in the $+\hat{y}$ direction, the area enclosed by the loop increases. If the applied magnetic field is along $+\hat{z}$, then the magnetic flux through the loop,

$$\Phi_B = \iint_S \vec{B} \cdot d\vec{S}, \tag{19}$$

increases because the area of the loop increases.

Hence an emf is induced. But the induced current does *not* assist the increase in flux. Instead, it flows in such a direction that the magnetic field produced by it, say $\vec{B}'$, is along $-\hat{z}$, thereby opposing the increase in the original magnetic flux.

Thus, for this case:

- rod moves in $+\hat{y}$ direction,
- enclosed area increases,
- flux due to external $\vec{B}$ increases,

- induced emf drives a **clockwise** current,
- this induced current produces $\vec{B}'$ toward $-\hat{z}$,
- hence the induced effect **opposes the change in flux**.

Lenz's Law

The direction of the induced emf/current is always such that the magnetic field produced by the induced current opposes the *change* in magnetic flux that caused it.

Therefore, Faraday's law must be written as

$$\mathcal{E} = -\frac{d\Phi_B}{dt}. \tag{20}$$

The negative sign is the mathematical expression of Lenz's law. It ensures that nature does not allow the induced current to reinforce the change that created it; instead, the induced response always resists that change.

Faraday's Law: Integral and Differential Forms

In the previous discussion, we stated Faraday's law as

$$\mathcal{E} = -\frac{d\Phi_B}{dt}. \tag{21}$$

Let us now make this statement more precise, first for the sliding-rod experiment and then in its general integral and differential forms.

Experiment 1 revisited: flux change due to change in enclosed area

Consider again the conducting rod sliding on rails in a uniform magnetic field. As the rod moves, the enclosed area of the loop changes, and hence the magnetic flux through the loop changes.

If the rod of length ℓ moves with speed $\vec{u}$ for a short time Δt, the additional vector area swept out is

$$\Delta\vec{S} = (\vec{\ell} \times \vec{u})\,\Delta t. \tag{22}$$

Hence the corresponding change in magnetic flux is

$$\Delta\Phi_B = \vec{B} \cdot \Delta\vec{S} = \vec{B} \cdot (\vec{\ell} \times \vec{u})\,\Delta t. \tag{23}$$

Using the scalar triple-product identity,

$$\vec{B} \cdot (\vec{\ell} \times \vec{u}) = \vec{\ell} \cdot (\vec{u} \times \vec{B}), \tag{24}$$

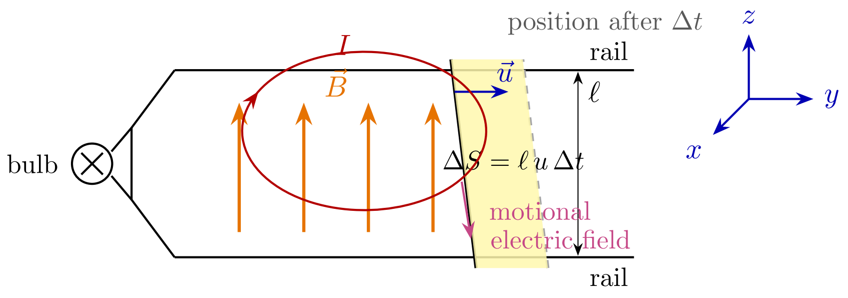


Figure 6: Sliding rod on rails: as the rod moves with speed $\vec{u}$, the enclosed area changes, causing the magnetic flux through the loop to change.

we obtain

$$\frac{d\Phi_B}{dt} = \vec{B} \cdot (\vec{\ell} \times \vec{u}) = \vec{\ell} \cdot (\vec{u} \times \vec{B}). \tag{25}$$

Thus the induced emf can be written as

$$\mathcal{E} = \vec{\ell} \cdot (\vec{u} \times \vec{B}). \tag{26}$$

Faraday's law in integral form (fixed circuit)

Since the induced emf drives current around a loop, we naturally describe it over a *closed path*. For a fixed closed contour C bounding a fixed surface S_c, Faraday's law is written as

$$\mathcal{E} = \oint_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \iint_{S_c} \vec{B} \cdot d\vec{S}. \tag{27}$$

If the surface S_c is fixed in time, then the time derivative can be taken inside the integral:

$$\oint_C \vec{E} \cdot d\vec{l} = -\iint_{S_c} \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}. \tag{28}$$

Faraday's Law (Integral Form)

$$\boxed{\oint_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \iint_{S_c} \vec{B} \cdot d\vec{S}}$$

This is the integral form of Faraday's law for a fixed contour. It is often called the **transformer emf** form.

Differential form using Stokes' theorem

By Stokes' theorem,

$$\oint_C \vec{E} \cdot d\vec{l} = \iint_{S_c} (\nabla \times \vec{E}) \cdot d\vec{S}. \tag{29}$$

Substituting this into Eq. (28), we get

$$\iint_{S_c} (\nabla \times \vec{E}) \cdot d\vec{S} = -\iint_{S_c} \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}. \tag{30}$$

Hence,

$$\iint_{S_c} \left(\nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} \right) \cdot d\vec{S} = 0. \tag{31}$$

Since this must hold for any arbitrary surface S_c, the integrands must be equal:

$$\boxed{\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}} \tag{32}$$

This is the **differential form of Faraday's law**.

Faraday's Law (Differential Form)

$$\boxed{\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}}$$

A time-varying magnetic field produces a circulating electric field.

Contrast with electrostatics

In electrostatics,

$$\nabla \times \vec{E} = 0, \tag{33}$$

which allows us to define

$$\vec{E} = -\nabla \phi. \tag{34}$$

But in electromagnetic induction,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \neq 0. \tag{35}$$

Therefore, the electric field is now *non-conservative*; in general, it can no longer be described purely as the gradient of a scalar potential.

Important Observation

In the presence of a time-varying magnetic field, the induced electric field forms closed loops. Hence it is fundamentally different from the electrostatic field.

General form for a moving circuit

The previous integral form was written for a *fixed* loop. More generally, if the circuit itself moves, the contour becomes time-dependent, say $C(t)$, and it encloses a time-dependent surface $S(t)$.

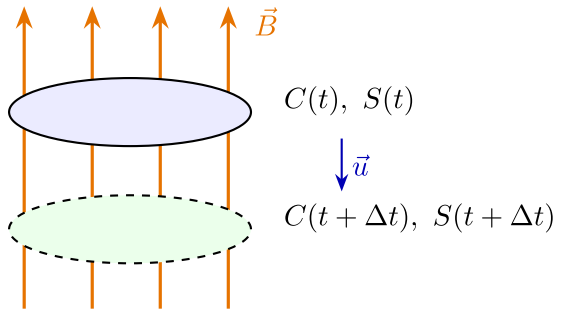


Figure 7: A moving closed loop in a magnetic field: even if $\vec{B}$ is not explicitly time-varying, the flux can change because the loop itself moves and the surface $S(t)$ changes with time.

For a moving circuit, the most complete form of Faraday's law is

$$\boxed{\mathcal{E} = \oint_{C(t)} (\vec{E} + \vec{u} \times \vec{B}) \cdot d\vec{l} = -\frac{d}{dt} \iint_{S(t)} \vec{B} \cdot d\vec{S}} \tag{36}$$

where $\vec{u}$ is the local velocity of the conducting element.

This is the general form that includes both:

- **transformer emf:** due to $\partial \vec{B}/\partial t$,
- **motional (generator) emf:** due to motion of the circuit in a magnetic field.

Final Summary

Faraday's law may appear in three closely related ways:

1. **Basic statement:**
$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

2. **Integral Maxwell form (fixed contour):**
$$\oint_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \iint_{S_c} \vec{B} \cdot d\vec{S}$$

3. **Differential Maxwell form:**
$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

For a moving circuit, the full emf is

$$\oint_{C(t)} (\vec{E} + \vec{u} \times \vec{B}) \cdot d\vec{l}.$$

Thus, a changing magnetic flux can arise either from a time-varying magnetic field or from motion of the circuit.

A Subtle Limitation of the Flux Rule

So far, we have used Faraday's law in the compact form

$$\mathcal{E} = -\frac{d\Phi_B}{dt}. \tag{37}$$

This is extremely useful, but it must be applied with care. The following thought experiment highlights an important subtlety.

Thought experiment: a "flux-rule paradox"

Consider the circuit shown in Fig. 8. The lower part of the circuit lies in a uniform magnetic field directed *into the page*. An ammeter is connected in the upper branch, and a switch can connect an additional vertical branch to the top wire.

If one uses the flux rule too casually, one may argue as follows:

- with the switch open, there is one loop,

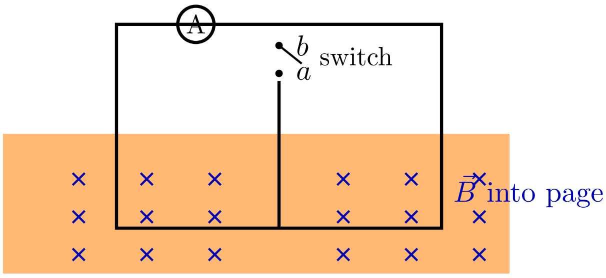


Figure 8: Thought experiment illustrating a subtle limitation of the flux rule. The switch changes the circuit topology instantaneously.

- when the switch is closed, the conducting path is redefined,
- therefore the "flux linked with the circuit" appears to change suddenly,
- so one may incorrectly expect a current pulse in the ammeter.

However, the ammeter does *not* register a genuine induced current in this idealized argument.

Why is there no real induced current?

The reason is that the quantity $\frac{d\Phi_B}{dt}$ appearing in Eq. (37) must refer to the rate of change of flux through a *well-defined loop that evolves continuously in time.* In the present thought experiment, the switch changes the *topology* of the conducting path instantaneously. This is not the same as a continuous time variation of flux through a single smoothly evolving loop.

In other words, here the apparent "change in flux" is not caused by

1. a continuous change in the magnetic field, or
2. a continuous motion/deformation of a fixed circuit,

but by an abrupt redefinition of what we call the loop.

Hence, in this situation,

$$\frac{d\Phi_B}{dt}$$

does not represent a physically meaningful induction process in the usual sense. Therefore, there is no genuine emf spike predicted by Faraday's law from magnetic induction alone.

Important Concept

The flux rule must be applied only when the magnetic flux through a *continuously evolving circuit* changes with time. An instantaneous topological change of the circuit is *not* the same as a genuine time-rate of change of flux for a single loop.

Correct way to interpret the flux rule

The statement

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \tag{38}$$

is best understood as follows:

> *The emf around a circuit equals the negative rate of change of magnetic flux through that circuit, provided the circuit is a well-defined loop and the change occurs through a genuine physical time evolution.*

Thus, the flux rule is safely applicable in the following situations:

1. the magnetic field changes with time,
2. the circuit moves or deforms continuously,
3. or both happen simultaneously.

It is *not* to be used blindly when the circuit itself is redefined discontinuously by switching.

Boxed Summary

1. Faraday's law in flux form,
$$\mathcal{E} = -\frac{d\Phi_B}{dt},$$
assumes a **single, well-defined loop** evolving continuously in time.

2. If a switch instantaneously changes the circuit topology, the apparent change in flux is **not** a genuine physical $\dfrac{d\Phi_B}{dt}$ for one smooth loop.

3. Therefore, such a switching event does **not** by itself imply a real induced emf/current from Faraday induction.

4. The flux rule is applicable when flux changes because:

   (a) the magnetic field changes,

   (b) the circuit moves/deforms,

   (c) or both.

   (d) The flux rule
   $$\mathcal{E} = -\frac{d\Phi_B}{dt}$$
   does not by itself distinguish whether the flux change is caused by motion of the circuit or by a time-varying magnetic field. Physically, these correspond to:
   $$\oint (\vec{u} \times \vec{B}) \cdot d\vec{l} \quad \text{(motional emf)}$$
   and
   $$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{(transformer emf)}.$$

Case (a): Circuit Moves

Example 1: Sliding bar in a uniform magnetic field

Let us first consider the case in which the *circuit moves* in a static magnetic field. A conducting bar slides to the right with speed $\vec{u}$ on two fixed rails, forming a rectangular loop. The magnetic field is uniform and directed *out of the page*, as indicated by the blue dots.

We assume that the induced current is sufficiently small, so that the magnetic field produced by that current may be neglected. Hence the external magnetic field $\vec{B}$ may be taken as unchanged.

Let the distance between the two rails be w, and let the instantaneous width of the loop be $x(t)$. Then the

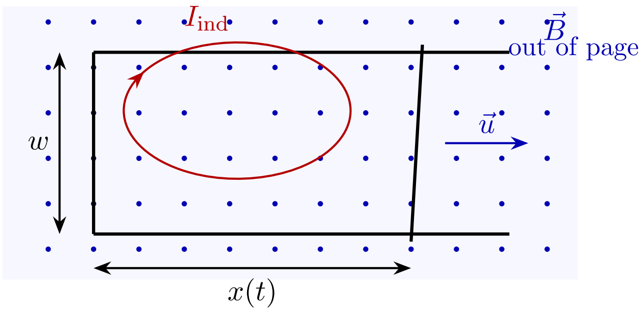


Figure 9: A sliding conducting bar forms a loop of changing area in a uniform magnetic field.

enclosed area is

$$A(t) = w\,x(t). \tag{39}$$

Since the magnetic field is uniform and perpendicular to the plane of the loop, the magnetic flux through the loop is

$$\Phi_B = \iint_S \vec{B}\cdot d\vec{S} = B\,A(t) = B\,w\,x(t). \tag{40}$$

Therefore, by the flux rule,

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\big(Bwx(t)\big). \tag{41}$$

Because B and w are constants,

$$\mathcal{E} = -Bw\frac{dx}{dt}. \tag{42}$$

If the bar moves to the right with speed u, then

$$\frac{dx}{dt} = u. \tag{43}$$

Hence,

$$\mathcal{E} = -Bwu. \tag{44}$$

So the *magnitude* of the induced emf is

$$\boxed{|\mathcal{E}| = Bwu.} \tag{45}$$

Direction of the induced current

As the bar moves to the right, the area of the loop increases. Hence the magnetic flux *out of the page* increases. By Lenz's law, the induced current must oppose this increase in flux. Therefore, it must produce a magnetic field *into the page*. Using the right-hand rule, the induced current is therefore **clockwise**.

Physical origin of the emf

Although the flux rule gives the correct emf immediately, the physical origin of the emf here is the magnetic force on charges inside the moving conductor.

A charge q inside the bar moves with the bar at velocity $\vec{u}$. Hence it experiences the magnetic force

$$\vec{F}_B = q(\vec{u} \times \vec{B}). \tag{46}$$

This acts like an effective electric field inside the moving bar:

$$\vec{E}_{\text{mot}} = \vec{u} \times \vec{B}. \tag{47}$$

Therefore, the emf may also be written as

$$\mathcal{E} = \oint_C (\vec{u} \times \vec{B}) \cdot d\vec{l}. \tag{48}$$

In this example, only the *moving bar* contributes, because the other parts of the circuit are stationary. Thus,

$$|\mathcal{E}| = \int_{\text{bar}} |\vec{u} \times \vec{B}|\, dl = uB \int_{\text{bar}} dl = uBw. \tag{49}$$

Hence again,

$$\boxed{|\mathcal{E}| = Bwu.} \tag{50}$$

So the flux rule and the motional-force picture give exactly the same result.

Boxed Summary

For a circuit moving in a static magnetic field:

1. the magnetic flux changes because the **area of the loop changes**,
2. the induced emf is given by
$$\mathcal{E} = -\frac{d\Phi_B}{dt},$$
3. for the sliding-bar example,
$$|\mathcal{E}| = Bwu,$$
4. the physical origin of this emf is the magnetic force on charges in the moving conductor:
$$\vec{F}_B = q(\vec{u} \times \vec{B}).$$

Thus, this is a standard example of **motional emf**.

(b) Circuit Moves: Example 2 - General derivation of motional emf

We now generalize the previous sliding-bar example to an *arbitrarily shaped loop* moving in a static magnetic field. This is the basic idea behind a **generator**. In contrast, in a **transformer**, the coil is fixed and the magnetic field changes with time.

Physical distinction

- **Generator / motional emf:** the circuit moves through a magnetic field.
- **Transformer emf:** the circuit is fixed, but the magnetic field changes with time.

Moving loop at times t and $t + \Delta t$

Consider a loop of wire at time t. After a short time Δt, the loop has moved to a new position. Let $S(t)$ be any surface bounded by the loop at time t, and let $S(t + \Delta t)$ be the corresponding surface bounded by the loop at time $t + \Delta t$.

The two positions of the loop are connected by a narrow *ribbon-like* surface generated by the motion of the wire elements.

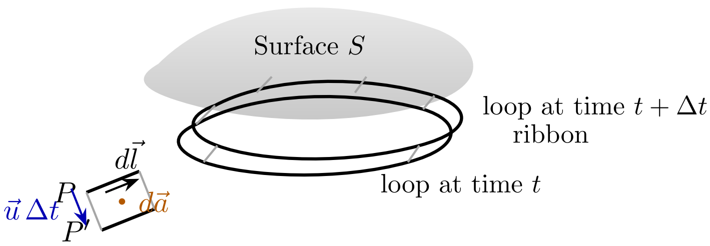


Figure 10: An arbitrary loop moving from time t to time $t + \Delta t$. The ribbon surface is swept out by the moving wire.

Change in flux through the moving loop

The magnetic flux through the loop at time t is

$$\Phi_B(t) = \iint_{S(t)} \vec{B} \cdot d\vec{S}. \tag{51}$$

Similarly, at time $t + \Delta t$,

$$\Phi_B(t + \Delta t) = \iint_{S(t+\Delta t)} \vec{B} \cdot d\vec{S}. \tag{52}$$

Hence the change in flux is

$$d\Phi_B = \Phi_B(t + \Delta t) - \Phi_B(t). \tag{53}$$

For a small displacement, this change is exactly the flux through the narrow ribbon swept out by the moving loop:

$$d\Phi_B = \iint_{\text{ribbon}} \vec{B} \cdot d\vec{S}. \tag{54}$$

Now consider a small wire element $d\vec{l}$. If that element moves with velocity $\vec{u}$, then in time dt it sweeps out a small area vector

$$d\vec{S} = (\vec{u} \times d\vec{l})\, dt. \tag{55}$$

Therefore,

$$d\Phi_B = \oint \vec{B} \cdot d\vec{S} = \oint \vec{B} \cdot (\vec{u} \times d\vec{l})\, dt. \tag{56}$$

Dividing by dt, we get

$$\frac{d\Phi_B}{dt} = \oint \vec{B} \cdot (\vec{u} \times d\vec{l}). \tag{57}$$

Using the scalar triple-product identity,

$$\vec{B} \cdot (\vec{u} \times d\vec{l}) = (\vec{B} \times \vec{u}) \cdot d\vec{l} = -(\vec{u} \times \vec{B}) \cdot d\vec{l}, \tag{58}$$

so that

$$\boxed{\frac{d\Phi_B}{dt} = -\oint (\vec{u} \times \vec{B}) \cdot d\vec{l}} \tag{59}$$

Connection with magnetic force on charge

A charge q in the moving wire experiences the magnetic force

$$\vec{F}_B = q(\vec{w} \times \vec{B}), \tag{60}$$

where $\vec{w}$ is the actual velocity of the charge.

Now the charge velocity may be written as

$$\vec{w} = \vec{u} + \vec{v}_{\parallel}, \tag{61}$$

where

- $\vec{u}$ is the velocity of the wire element itself,
- $\vec{v}_{\parallel}$ is the drift velocity of the charge along the wire.

Hence,

$$\vec{w} \times \vec{B} = \vec{u} \times \vec{B} + \vec{v}_{\parallel} \times \vec{B}. \tag{62}$$

But $\vec{v}_{\parallel}$ is along the wire, i.e. along $d\vec{l}$. Therefore,

$$(\vec{v}_{\parallel} \times \vec{B}) \cdot d\vec{l} = 0. \tag{63}$$

So, for the line integral around the circuit,

$$(\vec{w} \times \vec{B}) \cdot d\vec{l} = (\vec{u} \times \vec{B}) \cdot d\vec{l}. \tag{64}$$

Thus the magnetic force per unit charge contributes an emf

$$\mathcal{E} = \oint \frac{\vec{F}_B}{q} \cdot d\vec{l} = \oint (\vec{u} \times \vec{B}) \cdot d\vec{l}. \tag{65}$$

Using Eq. (59), we finally obtain

$$\boxed{\mathcal{E} = \oint (\vec{u} \times \vec{B}) \cdot d\vec{l} = -\frac{d\Phi_B}{dt}} \tag{66}$$

This is the general expression for **motional emf**.

Key Result: Motional emf

For an arbitrarily shaped loop moving in a static magnetic field,

$$\mathcal{E} = \oint (\vec{u} \times \vec{B}) \cdot d\vec{l} = -\frac{d\Phi_B}{dt}.$$

So, even when the magnetic field does *not* change with time, an emf is induced if the circuit moves so that the magnetic flux through it changes.

Boxed Summary

1. A moving loop sweeps out a narrow **ribbon surface**.
2. The change in magnetic flux is the flux through this ribbon.
3. For a small wire element,
$$d\vec{S} = (\vec{u} \times d\vec{l})\, dt.$$
4. Therefore,
$$\frac{d\Phi_B}{dt} = -\oint (\vec{u} \times \vec{B}) \cdot d\vec{l}.$$
5. Since $(\vec{u} \times \vec{B})$ is the magnetic-force-per-unit-charge contribution along the moving wire,
$$\mathcal{E} = \oint (\vec{u} \times \vec{B}) \cdot d\vec{l} = -\frac{d\Phi_B}{dt}.$$

This is the general derivation of **motional emf** and forms the basis of electric generators.

(c) Magnetic field changes; circuit is fixed

We now consider the complementary case to motional emf. Here, the **circuit is fixed**, but the **magnetic field changes with time**. This is the basic mechanism behind **transformer emf**.

Starting point: Faraday's law in differential form

For a time-varying magnetic field,
$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}. \tag{67}$$

Applying Stokes' theorem over a fixed surface S_c bounded by a fixed closed contour C, we get
$$\oint_C \vec{E} \cdot d\vec{l} = \iint_{S_c} (\nabla \times \vec{E}) \cdot d\vec{S}. \tag{68}$$

Substituting Eq. (67),
$$\oint_C \vec{E} \cdot d\vec{l} = -\iint_{S_c} \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}. \tag{69}$$

Since the contour C and surface S_c are fixed in time, the time derivative can be taken outside the surface integral:
$$\oint_C \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \iint_{S_c} \vec{B} \cdot d\vec{S}. \tag{70}$$

But

$$\Phi_B = \iint_{S_c} \vec{B} \cdot d\vec{S} \tag{71}$$

is the magnetic flux through the fixed surface S_c. Therefore,

$$\boxed{\oint_C \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}} \tag{72}$$

Thus, the induced emf is

$$\boxed{\mathcal{E}_{\text{induced}} = \oint_C \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}} \tag{73}$$

Important Point

In this case, the circuit does **not** move. The induced emf arises because a **time-varying magnetic field creates a circulating electric field**.

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

So here the emf is of **transformer type**, not motional type.

Physical meaning

In electrostatics,

$$\nabla \times \vec{E} = 0, \tag{74}$$

and hence the electric field can be written as

$$\vec{E} = -\nabla \phi. \tag{75}$$

But here,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \neq 0. \tag{76}$$

So the induced electric field is **non-conservative**. Its field lines form closed loops, and hence

$$\oint_C \vec{E} \cdot d\vec{l} \neq 0. \tag{77}$$

That non-zero circulation is exactly the induced emf around the fixed circuit.

Example: Transformer

A standard example is a transformer. The secondary coil is fixed, but the current in the primary coil changes with time. This produces a time-varying magnetic field, which links the secondary coil and induces an emf in it.

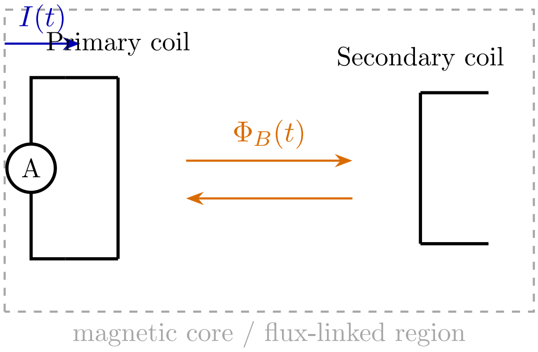


Figure 11: Transformer principle: the secondary coil is fixed, but a time-varying current in the primary produces a time-varying magnetic flux, inducing emf in the secondary.

If the magnetic flux through the secondary coil is $\Phi_B(t)$, then the induced emf in the secondary is

$$\mathcal{E}_s = -\frac{d\Phi_B}{dt}. \tag{78}$$

For a coil of N turns, this becomes

$$\boxed{\mathcal{E} = -N\frac{d\Phi_B}{dt}} \tag{79}$$

Comparison with the moving-circuit case

It is very important to distinguish the two physical mechanisms:

- **Circuit moves in magnetic field:**

  $$\mathcal{E} = \oint (\vec{u} \times \vec{B}) \cdot d\vec{l}$$

  Here the emf arises from the magnetic force on charges in the moving conductor.

- **Circuit fixed, magnetic field changes:**

  $$\mathcal{E} = \oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

  Here the emf arises because a time-varying magnetic field creates a circulating electric field.

Although both are often written compactly using the flux rule, the *physical origins* are different.

Key Takeaway

From the flux rule alone,

$$\mathcal{E} = -\frac{d\Phi_B}{dt},$$

one cannot always distinguish whether the flux changes because

1. the circuit moves, or
2. the magnetic field changes with time.

But physically, the two situations are different:

- **Moving circuit:**
$$\mathcal{E} = \oint (\vec{u} \times \vec{B}) \cdot d\vec{l}$$
- **Changing field:**
$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

So the flux rule gives the **net emf**, while the underlying physical explanation depends on the situation.

Energy Stored in a Magnetic Field

In electrostatics, we calculated the work required to build up a charge distribution, and that work appeared as the energy stored in the electric field. In magnetostatics, the analogous idea is slightly different: to establish a magnetic field, we must build up a *current distribution*. So the relevant question is: *who does the work while the current is being established?*

Does the magnetic field do work on a charge?

Suppose a charge q moves with velocity $\vec{v}$ and undergoes a small displacement $d\vec{l}$.

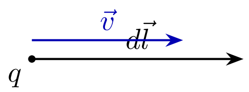


Figure 12: A charge q moving with velocity $\vec{v}$ through a small displacement $d\vec{l}$.

The total Lorentz force on the charge is

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B}). \tag{80}$$

Hence the small work done on the charge is

$$dW = \vec{F} \cdot d\vec{l} = \left[q\vec{E} + q(\vec{v} \times \vec{B})\right] \cdot d\vec{l}. \tag{81}$$

Now, since

$$\vec{v} = \frac{d\vec{l}}{dt}, \tag{82}$$

we may write

$$dW = \left[q\vec{E} + q\left(\frac{d\vec{l}}{dt} \times \vec{B}\right)\right] \cdot d\vec{l}. \tag{83}$$

Therefore,

$$dW = q\vec{E} \cdot d\vec{l} + q\left(\frac{d\vec{l}}{dt} \times \vec{B}\right) \cdot d\vec{l}. \tag{84}$$

Using the scalar triple-product identity,

$$\left(\frac{d\vec{l}}{dt} \times \vec{B}\right) \cdot d\vec{l} = \vec{B} \cdot \left(d\vec{l} \times \frac{d\vec{l}}{dt}\right). \tag{85}$$

But $d\vec{l}$ and $\frac{d\vec{l}}{dt}$ are parallel to each other, so

$$d\vec{l} \times \frac{d\vec{l}}{dt} = 0. \tag{86}$$

Hence,

$$\left(\frac{d\vec{l}}{dt} \times \vec{B}\right) \cdot d\vec{l} = 0. \tag{87}$$

So the work done reduces to

$$\boxed{dW = q\vec{E} \cdot d\vec{l}} \tag{88}$$

Important conclusion

Thus, the magnetic part of the Lorentz force does *no work* on a moving charge:

$$\boxed{q(\vec{v} \times \vec{B}) \cdot d\vec{l} = 0} \tag{89}$$

This is because the magnetic force is always perpendicular to the instantaneous velocity of the charge.

Key Physical Point

A magnetic field can change the *direction* of motion of a charge, but it cannot change its *speed*. Therefore, the magnetic field by itself does **no work** on a charge.

Then how is magnetic energy stored?

If the magnetic field itself does no work, then the energy stored in a magnetic field cannot come directly from magnetic force. Instead, when a current is built up in a circuit, the required work is done by the *electric field* associated with the source.

This is analogous to electrostatics, where the stored energy in a capacitor is obtained from the work done in assembling charge:

$$U_E = \frac{1}{2}CV^2. \tag{90}$$

Similarly, in magnetic systems, energy is stored while establishing current, and that work is supplied externally. This will lead us to the expression for magnetic energy stored in an inductor in the next step.

Boxed Summary

1. The Lorentz force on a charge is
$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B}).$$
2. The work done on the charge is
$$dW = \vec{F} \cdot d\vec{l}.$$
3. Since $(\vec{v} \times \vec{B})$ is perpendicular to $\vec{v}$,
$$q(\vec{v} \times \vec{B}) \cdot d\vec{l} = 0.$$
4. Therefore, the magnetic field does **no work** on a charge.
5. The energy stored in magnetic systems is supplied by the **electric field / external source** while building up the current.

Energy Stored in a Magnetic Field

A natural question now is: *is there energy stored in the magnetic field?*

In electrostatics, we found the energy stored in the electric field by calculating the work required to build up a charge distribution. In magnetostatics, the analogous idea is to calculate the work required to build up a *current distribution.*

From the previous discussion, we already know that the magnetic field itself does *no work* on a charge, because

$$q(\vec{v} \times \vec{B}) \cdot d\vec{l} = 0.$$

So if magnetic energy is stored, it cannot come from direct work done by $\vec{B}$. The correct explanation comes from **Faraday's law**.

Building up the current gradually

Suppose the final current density we want to establish is $\vec{J}_0(\vec{r})$. The corresponding magnetic field is $\vec{B}_0(\vec{r})$, given by the Biot–Savart law:

$$\vec{B}_0(\vec{r}) = \frac{\mu_0}{4\pi} \iiint_V \frac{\vec{J}_0(\vec{r}\,') \times \hat{\mathbf{R}}}{R^2}\, dv', \qquad \mathbf{R} = \vec{r} - \vec{r}\,'. \tag{91}$$

Because the Biot–Savart law is linear in $\vec{J}$, if we scale the current density by a factor α, the magnetic field also scales by the same factor:

$$\vec{J}(\vec{r}, t) = \alpha(t)\, \vec{J}_0(\vec{r}) \quad \Longrightarrow \quad \vec{B}(\vec{r}, t) = \alpha(t)\, \vec{B}_0(\vec{r}). \tag{92}$$

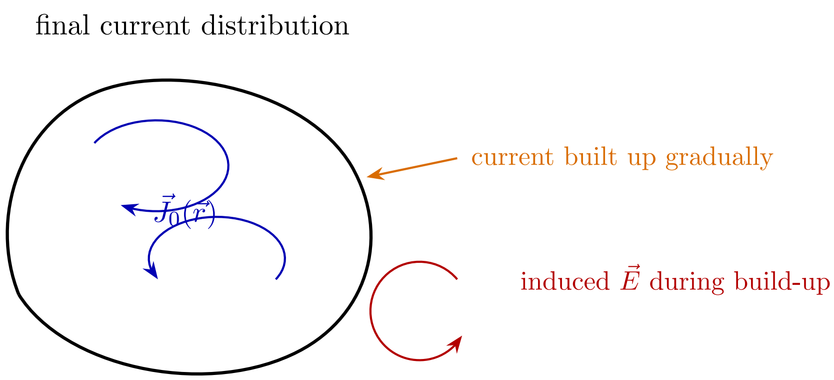


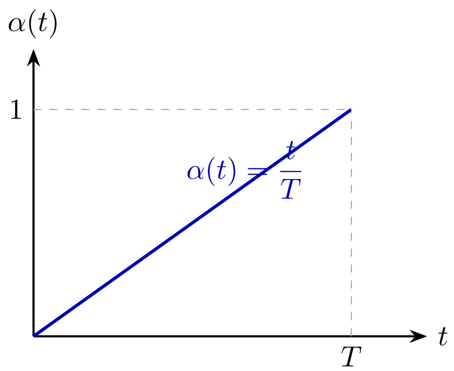
Figure 13: A current distribution is built gradually from zero to its final value $\vec{J}_0(\vec{r})$. During this process, the magnetic field changes with time, and hence an induced electric field appears.

To make the algebra simple, let us assume that the buildup is *slow* and *linear in time*:

$$\alpha(t) = \frac{t}{T}, \qquad 0 \le t \le T. \tag{93}$$

Then

$$\vec{J}(\vec{r}, t) = \frac{t}{T}\,\vec{J}_0(\vec{r}), \qquad \vec{B}(\vec{r}, t) = \frac{t}{T}\,\vec{B}_0(\vec{r}). \tag{94}$$



$$\vec{J}(\vec{r}, t) = \alpha(t)\,\vec{J}_0(\vec{r})$$
$$\vec{B}(\vec{r}, t) = \alpha(t)\,\vec{B}_0(\vec{r})$$

Figure 14: A simple linear buildup of current and magnetic field from zero to their final values.

Therefore,

$$\frac{\partial \vec{B}}{\partial t} = \frac{1}{T}\,\vec{B}_0(\vec{r}), \tag{95}$$

and hence, from Faraday's law,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\frac{1}{T}\,\vec{B}_0(\vec{r}) \neq 0. \tag{96}$$

This is the key point: while the magnetic field is being built up, a non-conservative electric field is induced. That induced electric field can do work.

Power supplied during buildup

We know that if a current density $\vec{J}$ exists in an electric field $\vec{E}$, then the power delivered by the field per unit volume is

$$p = \vec{J} \cdot \vec{E}. \tag{97}$$

Side Note: Why $\vec{J} \cdot \vec{E}$ is work done per unit volume per unit time

Consider a small volume element dv containing charge density ρ_v. If the charges inside it move with drift velocity $\vec{u}$, then the current density is

$$\vec{J} = \rho_v \vec{u}. \tag{98}$$

Now, the electric force on a small charge element dq is

$$d\vec{F} = dq\, \vec{E}. \tag{99}$$

If this charge moves with velocity $\vec{u}$, then the rate at which work is done on it is

$$\frac{dW}{dt} = \vec{F} \cdot \vec{u}. \tag{100}$$

So, for the charge element dq,

$$\frac{dW}{dt} = (dq\, \vec{E}) \cdot \vec{u} = dq\, (\vec{E} \cdot \vec{u}). \tag{101}$$

Dividing by the volume element dv,

$$\frac{1}{dv} \frac{dW}{dt} = \frac{dq}{dv}\, \vec{E} \cdot \vec{u}. \tag{102}$$

But

$$\frac{dq}{dv} = \rho_v, \tag{103}$$

hence

$$\frac{1}{dv} \frac{dW}{dt} = \rho_v\, \vec{E} \cdot \vec{u}. \tag{104}$$

Using $\vec{J} = \rho_v \vec{u}$, we get

$$\boxed{\vec{J} \cdot \vec{E} = \frac{\text{work done}}{\text{unit volume} \times \text{unit time}}} \tag{105}$$

Therefore, $\vec{J} \cdot \vec{E}$ is the **power density** delivered by the electric field to charges.

Units check:

$$\vec{J} \cdot \vec{E} \sim \left(\frac{\mathrm{A}}{\mathrm{m}^2}\right) \left(\frac{\mathrm{V}}{\mathrm{m}}\right) = \frac{\mathrm{W}}{\mathrm{m}^3},$$

which is indeed power per unit volume.

However, during the buildup of current, the induced electric field opposes the increase of current (Lenz's law). Therefore, the *external agent* must supply power density

$$p_{\text{ext}} = -\vec{J} \cdot \vec{E}. \tag{106}$$

In an ideal lossless buildup, this externally supplied power is what gets stored as magnetic energy. Hence the total magnetic energy stored is

$$U_B = \int_0^T \left[- \iiint_V \vec{J} \cdot \vec{E} \, dv \right] dt. \tag{107}$$

Now, for the slowly varying case, we use

$$\vec{J} = \nabla \times \vec{H}. \tag{108}$$

Substituting into Eq. (107),

$$U_B = -\int_0^T \iiint_V (\nabla \times \vec{H}) \cdot \vec{E} \, dv \, dt. \tag{109}$$

Using the vector identity

$$\nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H}), \tag{110}$$

we obtain

$$-\vec{E} \cdot (\nabla \times \vec{H}) = \nabla \cdot (\vec{E} \times \vec{H}) - \vec{H} \cdot (\nabla \times \vec{E}). \tag{111}$$

Therefore,

$$U_B = \int_0^T \iiint_V \left[\nabla \cdot (\vec{E} \times \vec{H}) - \vec{H} \cdot (\nabla \times \vec{E}) \right] dv \, dt. \tag{112}$$

So,

$$U_B = \int_0^T \left[\iiint_V \nabla \cdot (\vec{E} \times \vec{H}) \, dv \right] dt - \int_0^T \left[\iiint_V \vec{H} \cdot (\nabla \times \vec{E}) \, dv \right] dt. \tag{113}$$

Using the divergence theorem,

$$\iiint_V \nabla \cdot (\vec{E} \times \vec{H}) \, dv = \oiint_S (\vec{E} \times \vec{H}) \cdot d\vec{a}. \tag{114}$$

For localized sources, if we take the bounding surface far away, this surface integral vanishes.[1]

Hence,

$$U_B = -\int_0^T \iiint_V \vec{H} \cdot (\nabla \times \vec{E}) \, dv \, dt. \tag{115}$$

Now use Faraday's law:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}. \tag{116}$$

Therefore,

$$U_B = \int_0^T \iiint_V \vec{H} \cdot \frac{\partial \vec{B}}{\partial t} \, dv \, dt. \tag{117}$$

[1]Physically, in the quasistatic buildup considered here, we neglect radiation losses, so there is no net Poynting flux escaping to infinity.

Interchanging the order of integration,

$$U_B = \iiint_V \left[\int_0^T \vec{H} \cdot \frac{\partial \vec{B}}{\partial t}\, dt \right] dv. \tag{118}$$

Since $d\vec{B} = \dfrac{\partial \vec{B}}{\partial t}\, dt$, we may write

$$\boxed{U_B = \iiint_V \left[\int_{\vec{0}}^{\vec{B}} \vec{H} \cdot d\vec{B} \right] dv} \tag{119}$$

This is the **general expression** for magnetic energy stored in a medium.

Energy density for a linear magnetic medium

If the medium is linear, then

$$\vec{B} = \mu \vec{H} \qquad \Longrightarrow \qquad \vec{H} = \frac{\vec{B}}{\mu}. \tag{120}$$

Hence the magnetic energy density is

$$u_B = \int_{\vec{0}}^{\vec{B}} \vec{H} \cdot d\vec{B} = \int_{\vec{0}}^{\vec{B}} \frac{\vec{B}}{\mu} \cdot d\vec{B}. \tag{121}$$

For a fixed field direction, this becomes

$$u_B = \int_0^B \frac{B'}{\mu}\, dB' = \frac{1}{\mu} \int_0^B B'\, dB' = \frac{B^2}{2\mu}. \tag{122}$$

Therefore,

$$\boxed{u_B = \frac{B^2}{2\mu}} \tag{123}$$

and the total stored magnetic energy is

$$\boxed{U_B = \iiint_V \frac{B^2}{2\mu}\, dv} \tag{124}$$

In free space,

$$\boxed{u_B = \frac{B^2}{2\mu_0}} \tag{125}$$

Physical interpretation

This result is very important conceptually:

- The magnetic field itself does **no work** on charges.
- But while the magnetic field is being established, $\frac{\partial \vec{B}}{\partial t} \neq 0$.
- Hence, by Faraday's law, an electric field is induced:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}.$$

- That induced electric field can do work.
- The external source must overcome this induced field in order to build up the current.
- The work supplied in this process gets stored as magnetic-field energy.

Thus:

> *Although a static magnetic field does no work, the process of building up the magnetic field requires work, and that work is stored in the magnetic field.*

Key Idea

At the beginning, before the current is established, there is no magnetic field. At the end, after the current becomes steady, there is a static magnetic field but no induced electric field. However, *during the buildup*, $\partial \vec{B}/\partial t \neq 0$, so an induced electric field exists, and that is what accounts for the work needed to store energy in the magnetic field.

Boxed Summary

1. To find magnetic energy, we imagine building the current distribution gradually from zero.
2. Because $\vec{B}$ changes during this process,
$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \neq 0,$$
so an induced electric field appears.
3. The power density supplied by the external source is
$$p_{\text{ext}} = -\vec{J} \cdot \vec{E}.$$
4. Hence the total magnetic energy stored is
$$U_B = \iiint_V \left[\int_{\vec{0}}^{\vec{B}} \vec{H} \cdot d\vec{B} \right] dv.$$
5. For a linear medium ($\vec{B} = \mu \vec{H}$),
$$u_B = \frac{B^2}{2\mu}, \qquad U_B = \iiint_V \frac{B^2}{2\mu}\, dv.$$
6. In free space,
$$u_B = \frac{B^2}{2\mu_0}.$$